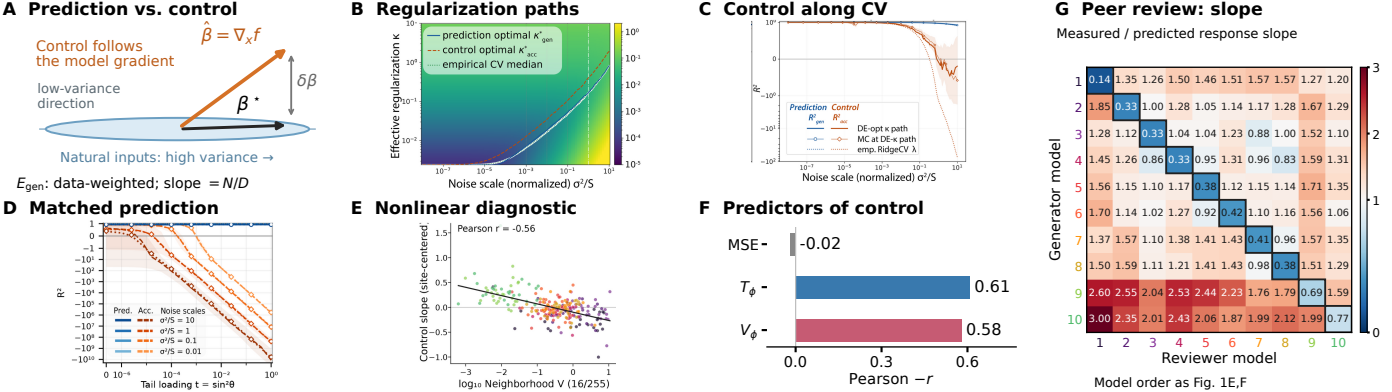

A Prediction vs. control
$\hat{\beta} = \nabla_x f$
Control follows the model gradient
low-variance direction
$\delta\beta$
$\beta^\star$
Natural inputs: high variance →
E_{gen}: data-weighted; slope $= N/D$
B Regularization paths
prediction optimal κ^*_{gen}
control optimal κ^*_{acc}
empirical CV median
Effective regularization κ
Noise scale (normalized) σ²/S
C Control along CV
Prediction R^2_{gen}
Control R^2_{acc}
DE-opt κ path
MC at DE-κ path
emp. RidgeCV λ
Noise scale (normalized) σ²/S
G Peer review: slope
Measured / predicted response slope
Generator model
Reviewer model
Model order as Fig. 1E,F
D Matched prediction
Pred. Acc. Noise scales
σ²/S = 10
σ²/S = 1
σ²/S = 0.1
σ²/S = 0.01
R²
Tail loading t = sin²θ
E Nonlinear diagnostic
Pearson r = -0.56
Control slope (site-centered)
log₁₀ Neighborhood V (16/255)
F Predictors of control
MSE
T_ϕ
V_ϕ
-0.02
0.61
0.58
Pearson −r